

Code-Layout, Readability and Reusability

Date	26 March 2026
Project Name	Smart Lender

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper Python indentation is maintained.
2	Proper File Structure	Files and folders are logically organized	Yes	Project is organized into templates, static, model, and application files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive variable names are used for applicant and loan details.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are named based on their specific tasks.
5	Code Comments	Inline and block comments explain logic	Partial	Comments are provided for important sections only.

6	Modular Design	Code is split into reusable functions/modules	Yes	Flask routes and ML models are separated into modules.
7	No Redundant code	Duplicate or unused code is removed	Yes	Repeated code has been minimized.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation and exception handling are implemented.

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Loan Prediction Model	Python (XGBoost)	Loan prediction process	High
2	Flask Application	Flask	Backend request handling	High
3	HTML Templates	HTML/CSS/Bootstrap	User Interface	High
4	Data Preprocessing Module	Python (Pandas)	Data cleaning and prediction	Medium
5	Model Loading Module	Joblib / Pickle	Loading trained ML model	High

Overall Code Quality Assessment

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate frontend,

		backend, and ML components.
Readability	5	Code is easy to understand with meaningful names and proper formatting.
Reusability	5	Reusable modules for prediction, preprocessing, and Flask routes.
Documentation / Comments	4	Sufficient comments for important logic; additional comments can improve maintainability.
Overall Score	4.8/5	The project follows good coding practices and is maintainable, reusable, and easy to extend.